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A Knowledge-based Decision Support System for Sulfur Pricing

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Abstract

In order to improve the effectiveness of the decision-making of sulfur pricing at PG Plant, based on the existing price decision-making process and pricing method, the aided decision-making of application information system is developed by applying the knowledge discovery method, and provided technical support to resolve the sulfur pricing in semi-structured decision problems.

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1. Introduction

The Current Situation in China Sulfur Market Sulfur is mainly used for phosphate fertilizer production which accounts for about 60% of sulfur consumption in China. There are almost no natural sulfur ore in China, the sulfur production has affiliates to the oil production, for example, Sulfur is recovered from the petroleum refining and natural gas purification.

Forecast according to International Fertilizer Industry Association (IFA): to 2013, world sulfur production will achieve 63.7 million tons and world sulfur consumption will achieve 60.7 million tons. Global sulfur demand and supply will remain potential surplus of 3 million tons to 4 million tons for the coming years.

According to the forecast, the total demand of sulfur in China is expected to be 14 million tons in 2011. The supply of sulfur in China depends on import primarily. Since 1995, China's sulfur imports are

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growing very fast. The imports maintain the quantity of 10 million tons in recent years. China's sulfur import in 2008 was 8414.94kt, attaining a year-on-year fall of 12.77%. The imports in 2009 was 12167.9 kt, increasing 44.60% compared to 2008.China's sulfur imports in 2010 was about 10489.8 kt, reducing about 14% compared to last year. China's dependence on imported sulfur is over 50% in recent years and China is now the largest sulfur consumer and importer in the world. Over 40% of the sulfur produced around the world is sold to China. China's sulfur imports account for 70% of global sulfur trade volume and have an outsized effect on the international sulfur market.

On the other hand, China's sulfur production enterprises are mainly concentrated in the oil refining subsidiary companies of Sinopec and PetroChina. In recent years， as the increasing of China's processing sour crude, the operating of gas desulfurization project and the increasing of the new oil refining project, domestic sulfur production capacity greatly increased and has currently reached more than 5.5 million tons, actual annual production is close to 3.5 million tons. Especially the mine production of Sinopec phosgene in Sichuan, the annual plan is 2-2.5 million tons/year and it will make the self-sufficiency rate of the original sulfur in China by original about 14% increase to 41 percent above. According to forecasts, the domestic sulfur production will be nearly 400 million tons in 2011, it still expects that imports will be about 10 million tons and import dependence will still be high. China's sulfur demand is heavily dependent on imports and the situation is unlikely to change soon.

2. The Key Affecting Factors

Because of China's sulfur demand is heavily dependent on import, thus China's sulfur producing enterprises set the price of their own product mainly according to the price of imported sulfur, their sulfur production, market etc. Following the pricing principle basis "taking market as the direct, smooth sales as the premise, improving efficiency as the goal ", short-term pricing of sulfur should consider the related factors which include the following aspects:

2.1. International sulfur price and the amount of imported

China as the world's largest importer of sulfur depends highly on the import of sulfur. Imported sulfur account for more than half of the total national sulfur consumption. International sulfur price is taking a decisive impact on the sulfur price fluctuation in China. The 2008 international sulfur prices experience amazing skyrocketing and slumping, sulfur import price rose from 3350-3900 yuan/ t in the early of the year to the highest 5600 ~ 5900 yuan/ t, again down to 600 ~ 700 yuan/t by the end of the year, the biggest drop is more than 90%, like riding a rollercoaster, causing the sulfur imports of China reduced by 12.77% in 2008 compared to the same period, about specific 8414.94 kt. In 2009, the sulfur market is smooth, the imported amount in the whole year is 12167.9 kt, amount of imported compared with 2008 has swollen 44.60%; 2010 international sulfur prices rise successively, domestic sulfur market is out of recession, frequent changes in price, the sulfur imports is about 10489.8 kt, around 14 percent reduction compared to the same period.

2.2. International oil price

Sulfur as the by-products of oil refining and gas purification, its price was directly affected by the international price of crude oil fluctuations. The most typical evidence is, in 2008, the price of crude oil experiences giant fluctuation, the price reached the highest level of 145.29 dollars a barrel in early July, then affected by the financial crisis, price plunged, had fallen to 33.87 dollars a barrel, drop as much as

76.7%. Meanwhile, the price of sulfur has directly affected by international oil price, experience a rollercoaster situation of big rise and fall, and its biggest fall more than 90%.

2.3. Domestic sulfur supply and demand

About 70 percent of our domestic sulfur consumption concentrates on phosphate production, so sulfur downstream demand depends mainly on phosphate market situation. In 2011, China will increase investment on agricultural production and sulfur domestic demand will increase correspondingly. On the supply side, as the new production capability of domestic sulfur is improved in 2010, self-sufficiency rate of domestic sulfur will be increased from 14% to over 41%. In the next few years, Chinese sulfur channel chain will occur major changes, and impact the sulfur imports. On the relatively stable demand circumstances, there will be a price war. It is expected to see that the trend of sulfur prices will decline.

2.4. Target market demand

The sulfur demand of the areas which are the main domestic consumer market including the south-west and Hubei area and so on is about (600 ~ 700) x 104t. PG's Gas field because of its remoteness, its relatively small sales areas which are mainly related to the dozens of manufacturers of Chongqing, Sichuan, Guizhou and Hubei and other regions, therefore the change of target market demand can directly influence the pricing decision of PG's sulfur.

2.5. Inventory of PG's sulfur

Sulfur pricing decisions must also consider the particularity of its production. Compared with other chemical products, sulfur production is attached to oil production, its production is involuntary and its supplement is rigid. Therefore, the main sales decisions of PG's sulfur use its own inventory levels as the important reference index other than to pursue maximum profits. Its pricing principle is taking into account benefits further under the premise of smooth sales.

2.6. The logistics cost

"Long-term value of sulfur focus on the sum of transportation and processing fee fluctuated because of sulfur is a by-product of oil production." As the international crude oil prices nose-dived in the second half of 2008, sea freight also fell dramatically. There is a huge decline in sea freight from Canada to China, for example, 66kt tons of freight has fallen to year-end $10 / t from $60 / t. Similarly, in mid-November 2010, diesel shortage in Sichuan and Chongqing led to the declination of transport capacity, rising logistics cost, stuck of transport, causing sulfur inventory of PG's gas field increased.

2.7. Policy factors

Policy factors can also affect the pricing decisions of sulfur. Government controls the export tariff of downstream fertilizer market and sulfur import VAT and so on to affect the market trend. Since 2010 January 1, China has resumed imposing the sulfur import tariffs, with the result that the sulfur import price is increased and the import intensity of sulfur is restrained to a certain degree.

According to the relevant reports: in order to improve the competitiveness of domestic sulfur manufacturer, realize uniform allocation of state resources and promote the improvement of China's sulfur industry systems, Sinopec will introduce unified purchase and sale policy refers to the sulfur and

paraffin of its affiliated companies. Currently, Sinopec has 27 sulfur producers; the market share is about 18%. The measure mentioned above will affect domestic sulfur market structure.

3. Architecture of the Proposed Knowledge-Based Dss

3.1. The Existing Pricing Processes

After the PG Plant starts operating, the sulfur it produces will replace part of the share of imports of sulfur, therefore the pricing of PG's sulfur should focus on referring to sulfur import prices, in principle, the DDP of PG's sulfur customer is roughly equivalent to the DDP of imported sulfur. PG's gas field has formulated a rational and effective price decision-making mechanism called "crude oil pricing method", using average sales price of this month as the guide price of next month. The pricing decision-making process of PG's gas field is shown in figure 1.

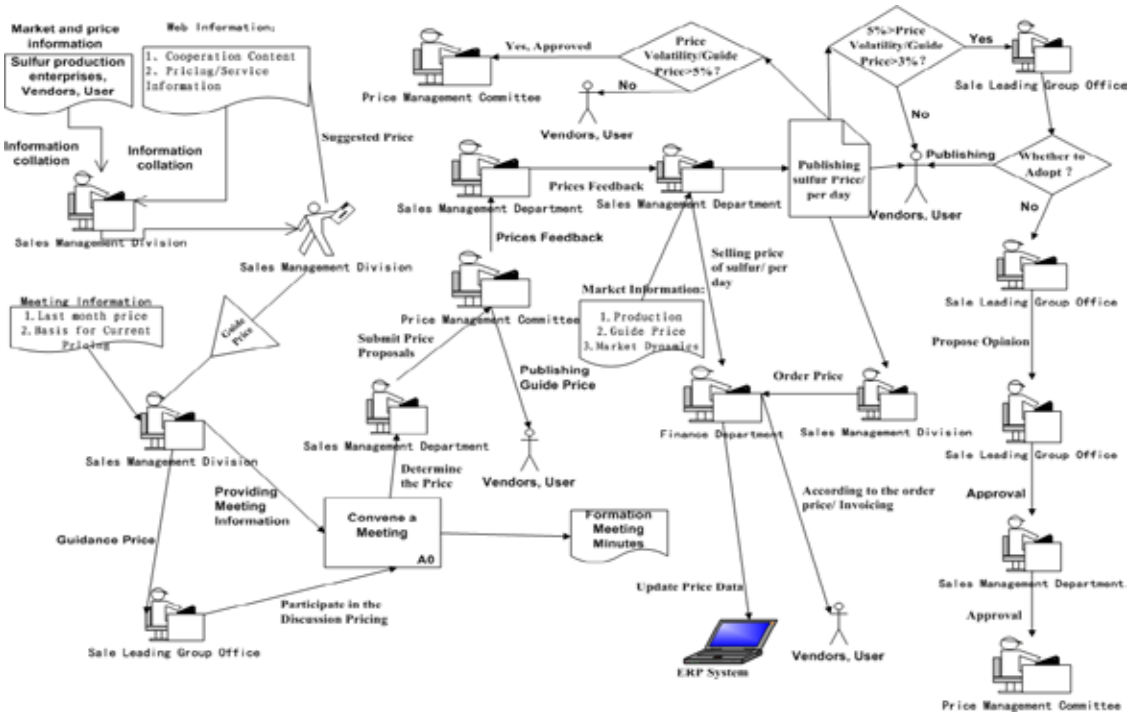

Fig.1. Pricing decision-making process of PG's gas field

3.2. Knowledge-Based Decision Support System Framework

In order to improve the effectiveness of the decisions of sulfur pricing at PG, we introduced the aided decision-making of application information system based on the existing price decision-making process and pricing method, and established a multi-objective decision-making model, helping to solve the problem of PG's sulfur pricing semi-structured decision. The system can query and analyze the information like supply and demand of the sulfur market, production inventory, sales, sulfur price fluctuations through the collection of sulfur historical data and the real-time data. The system can also

make a sulfur-pricing schemes report using decision support techniques. The frame structure and system functions of the system are shown in figure 2.

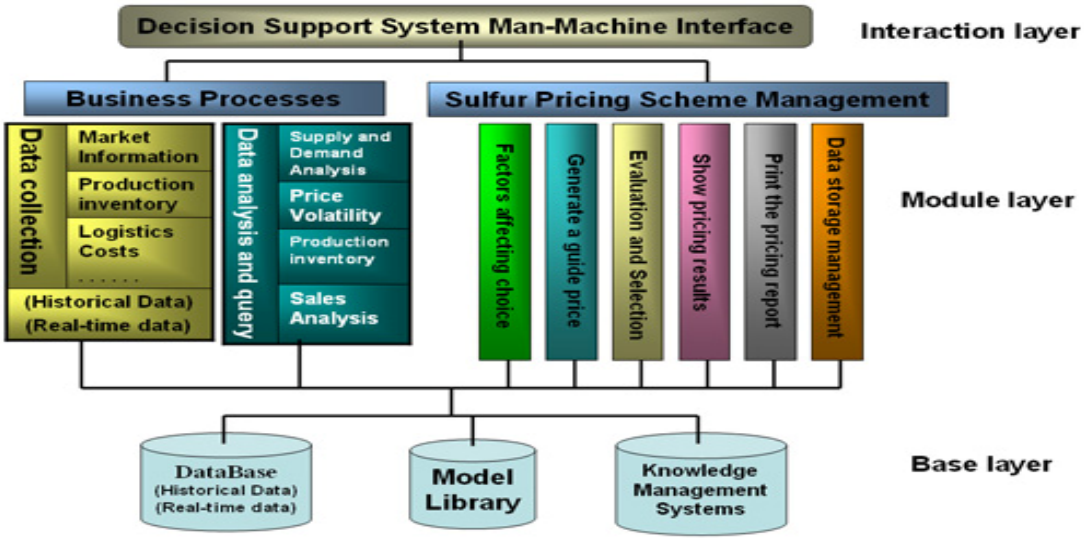

Fig.2. Information system framework and system functions

Anyhow, PG's sulfur production occupies a large proportion in domestic total sulfur production. The pricing decision of PG's sulfur are influenced by various factors, so we need to get the data information such as foreign and domestic sulfur market dynamic, sulfur price of other manufacturers, sulfur output and so on during the pricing process . The price formulation or substantial price adjustment needs strict process to ensure the implementation of the price. The introduction of computer information system supporting the decision-making and the application of the information technology with information processing and data analysis ability can make great contribution to improve the effectiveness of PG's sulfur pricing decisions and achieve the purposes of scientific decision-making

3.3. The Rule of Sulfur Pricing

PG's Gas Field is located in Sichuan Province, which has high concentrations of hydrogen sulfide(the average content of hydrogen sulfide is 14.6%).The natural gas produced by PG's Gas Field can be transported after purification treatment. With the purified natural gas, the annual production of the by-product sulfur is about 200 million tons which occupies countrywide 1/2. PG's Gas Field will become Asia's largest sulfur production base.

As its remote location, bad traffic conditions, and relatively small sales area, the sulfur is mainly sold to Chongqing, Sichuan, Guizhou, Hubei and other areas with dozens of manufacturers. The main transportation there is motor transport, rail transport, and shipping.

Sulfur is the by-product of the natural gas in PG's Gas field, the focus of natural gas purification is to ensure the normal natural gas production through desulfurization. Due to the high yield of PG's sulfur, only the sulfur is transported outside, can the purification plant continue to purify the gas, otherwise the PG's gas field can't maintain sustainable production and this will bring tremendous influence to the national economy and social development of the Yangtze River delta region. The smooth transportation

of sulfur is an important precondition for the normal production in PG's gas field. Therefore, the marketing strategy of PG's sulfur is to make the sales plan according to the market demand, based on maintaining reasonable inventory level and not affecting the normal natural gas production.

Sulfur as an important mineral resource related to the national economy and people's daily life, the formulation of its price needs to follow some rules. Generally speaking, the pricing target of their products is subordinate to the operation objectives of the enterprise. PG's gas field belongs to Sinopec, which means except considering the interests of investors and enterprise itself, it should also consider the interests of the state and society.

The price decision should be made according to the requirements of the company's target market strategy and market positioning strategy.

In summary, the pricing principle of PG's sulfur can be summarized as follows: taking market as the direct, smooth sales for the premise, improving efficiency as the goal. In another word, making sure the sales smooth firstly and then taking into account benefits. In the small market, we should pay more attention to big customers, and also the dealers. When others raise the price we should follow them, but if the product sold bad we should cut the price firstly.

In order to achieve the above principles, we establish the multi-objective decision-making model. Firstly an initial weight factor table is designed, which include international sulfur price, international oil price, port sulfur price, domestic supply, domestic demand, target market needs, domestic competitors, PG's sulfur production, PG's inventory of sulfur, and domestic logistics costs. All these weight factors are determined by expert evaluation method. While the weight factors are not the same when up or down, so different values are given.

In order to provide an effective decision based on historical cases, a weight factor table of pre-program is build. While a weight factor updating mechanism is also established by applying the knowledge discovery method.

4. Conclusions

As China's sulfur demand is heavily depending on imported sulfur, the pricing strategies of sulfur producing enterprises should mainly reference the price of imported sulfur, and many other affective factors. So it is a useful approach to apply the knowledge discovery method and develop the aided decision-making application information system, in order to improve the effectiveness of the sulfur pricing at PG.

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